

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash

Tools/Frameworks: PyTorch, TensorFlow, LangChain, OpenAI API, Streamlit, Docker, FastAPI, Kubernetes, AWS SageMaker, Pandas, TensorRT, Flask, HTML, CSS

Competence: Large Language Models (LLMs), Agentic AI, Machine Learning, MLOps, Deep Learning, Data Engineering, AI Model Optimization, CI/CD, Scalable Machine Learning, Data Visualization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, achieving improved model performance and training efficiency.
- Developed physics-informed hybrid loss functions to enhance model accuracy and stability for AI applications in metrology.
- Collaborated with PhD researchers to co-author a publication for the FLOMEKO conference.
- Built interactive dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and reduced processing time by 5x using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system using TensorRT, reducing compute costs and inference time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Created POCs leveraging LLM frameworks, YOLO, few-shot learning, and CNNs for client demonstrations.
- Conducted educational sessions on machine learning and big data for university students.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website offering AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and LLM frameworks for seamless user interaction.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for both structured and unstructured data ingestion.
- Automated data cleaning workflows, enabling efficient analytics for client use cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Implemented real-time forecasting models using time-series analysis and machine learning techniques for cryptocurrency price prediction.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Pioneered the development of the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specializing in Large Language Models (LLMs), Agentic AI, and advanced machine learning techniques.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Completed coursework in machine learning, data structures, and artificial intelligence.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python